

ALIF ERIKSANDI AGUSTINO

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EDUCATION

Brawijaya University
Computer Engineering Bachelors
GPA: 3.48

Malang City, Indonesia
August 2023 - Present

EXPERIENCE

AI Talent Factory (AITF) KOMDIGI x UB | AI Engineer

Malang, Indonesia | February 2026 - June 2026

Developing an AI chatbot for high school mentoring designed to adapt its conversational style to a high school level. Instead of providing direct answers, the chatbot employs pedagogical methods to foster two-way dialogue that encourages a growth mindset.

ROBOTIIK | Research and Development

Malang City, Indonesia | Mar 2025 - Dec 2025

- Focused on the electrical engineering and control systems for specialized robotics projects, including Search and Rescue (SAR) robots and Sumobots.
- Designed electrical schematics using Fritzing and executed precise component wiring and soldering to ensure reliable hardware connections.
- Identified technical requirements and selected optimal components (sensors, actuators, microcontrollers) based on specific robot specifications.

Google Developer Groups (GDGs) on Campus | Machine Learning

Malang City, Indonesia | Nov 2025 - Present

- Conducted Exploratory Data Analysis (EDA) using advanced filtering, grouping, and visualization techniques to uncover data patterns.
- Executed data preprocessing pipelines to convert raw data into clean, ML-ready formats for model training.

Google Developer Groups (GDGs) on Campus | Web Development

Malang City, Indonesia | Nov 2025 - Present

- Developed a comprehensive personal portfolio website to showcase projects and skills.
- Acquired practical skills in creating dynamic web experiences leveraging TypeScript and Next.js.

SKILLS

Programming Languages:

Libraries/Frameworks:

Tools / Platforms:

Databases:

Python, C, C++, HTML, CSS, JavaScript
Typescript, Next.js, Marker, Unsloth, PyTorch, Pandas
Visual Studio Code, MATLAB, Arduino IDE, Grafana, Fritzing, Atmel Studio,
Google Colab, Wandb, Gradio, Github, HuggingFace
SQL, MongoDB, InfluxDB

PROJECTS / OPEN-SOURCE

AI Chatbot for Mentoring Student

Python for Training Base Model LLM, Github, HuggingFace

- Engineered end-to-end data pipelines, including data crawling, extraction, cleansing, and Exploratory Data Analysis (EDA) to ensure high-quality training inputs.
- Curated and structured specialized datasets tailored for Continued Pre-Training (CPT), Supervised Fine-Tuning (SFT), Direct Preference Optimization (DPO), and Guardrails implementation.
- Fine-tuned and aligned a 9B-parameter base model (Qwen 3.5), developing a domain-specific LLM capable of answering student queries using pedagogical methods.

Air Quality Monitoring | Link

MQTT, Mosquito, PostgreSQL, and Grafana

- Developed an IoT-based monitoring system to track indoor air quality metrics using Python and MQTT.
- Implemented a Batch Processing (ETL) pipeline to extract sensor data, validate ranges, and load cleaned data into PostgreSQL.

- Containerized the application services using Docker and designed interactive Grafana dashboards to visualize CO2 and PM2.5 levels for office health analysis.

Integration of Reinforcement Learning Methods for Floor Cleaning Robot Optimization | [Link](#) *Python with Reinforcement Learning (Q-Learning)*

- Developed an autonomous robot simulation using Python and Reinforcement Learning (Q-Learning) to optimize cleaning paths and battery management.
- Implemented a reward-based decision-making system, allowing the agent to learn navigation and resource allocation strategies through trial and error.
- Built an interactive GUI using Pygame for real-time simulation and visualized training performance metrics (learning curves) using Matplotlib.

Food Recommendation System at Filkom UB Canteen Using Naive Bayes Algorithm | [Link](#) *Python, Naive Bayes Algorithm*

- Developed a food recommendation system for the university canteen using Python and the Naive Bayes algorithm with Laplace Smoothing to handle zero probabilities.
- Implemented a probabilistic model to predict top-3 menu recommendations based on user preferences (budget, taste, texture, popularity), enhancing decision-making for customers.
- Designed a lightweight, dependency-free application using Python's standard libraries to process categorical data and calculate recommendation scores.

Filtering, Sorting, Grouping, and Graphic Visualization | [Link](#) *Python with Pandas and NumPy*

- Conducted comprehensive Exploratory Data Analysis (EDA) on a dataset of smartphone specifications using Python and Pandas to derive actionable business insights.
- Performed advanced data manipulation techniques, including filtering, sorting, and grouping, to identify market trends and product positioning strategies.
- Executed Feature Engineering to create new metrics (e.g., value-for-money score) and implemented logic for anomaly detection to isolate data outliers.

Obstacle Following Robot | [Link](#) *Ultrasonic Sensor with Arduino UNO R3 as Controller*

- Developed an autonomous robot using Arduino and an Ultrasonic Sensor (HC-SR04) capable of tracking and maintaining a safe distance (10-20cm) from moving objects.
- Implemented a Smart Scanning feature with anomaly detection logic: if the target is lost or moves abruptly, the robot automatically scans left and right to reacquire the object.
- Utilized PWM speed control on the L298N driver for smooth motor operation and programmed a state machine in C++ to handle real-time decision-making (Forward, Backward, Stop, Scan).

CERTIFICATIONS

- Data Science (Fresh Graduate Academy) - Digital Talent Scholarship.
- Fundamental of Deep Learning - AITF (Digital Talent) - Digital Talent Scholarship.
- Intermediate Deep Learning - AITF (Digital Talent Academy) - Digital Talent Scholarship.
- Mastery In Deep Learning - AITF (Digital Talent Academy) - Digital Talent Scholarship.
- LLM: Dari Teori Hingga Kode - AITF (Digital Talent Academy) - Digital Talent Scholarship.
- Rekayasa Aplikasi LLM Terapan - AITF - Digital Talent Scholarship.
- Python Basic Certificate - **HackerRank**
- Reinforcement Learning Onramp Self-Paced Training Course - **MathWorks**
- Multibody Simulation Onramp Self-Paced Training Course - **MathWorks**
- Simscape Onramp Self-Paced Training Course - **MathWorks**
- System Composer Onramp Self-Paced Training Course - **MathWorks**